

Question (1): A data scientist is told that the characteristic equation of a data matrix he works with has the property that the roots are all distinct and real. The data scientist does not know the values of the roots of the characteristic equation but assumes that:

(a) the matrix is bound to be invertible, and
(b) also diagonalizable. Is the data scientist justified in making these assumptions about the matrix? Give appropriate reasons for your answer. [5 Marks]

Please provide detailed step by step answer to this question so that even newbie can understand



Question Analysis:

The problem requires us to verify the validity of two assumptions about a matrix A whose characteristic equation has distinct and real roots:

1. The matrix is **invertible**.
2. The matrix is **diagonalizable**.

Let's go step by step to determine whether these assumptions are justified.

Step 1: Recall Definitions

Characteristic Equation:

The characteristic equation of a matrix $A \in \mathbb{R}^{n \times n}$ is given by:

$$\det(A - \lambda I) = 0,$$

where λ are the eigenvalues of A , and I is the identity matrix.

Eigenvalues:

The roots of the characteristic equation are the eigenvalues of A . In this question, the eigenvalues are given as **real** and **distinct**.

Invertibility of a Matrix:

A square matrix A is invertible if and only if $\det(A) \neq 0$. This means none of the eigenvalues of A can be zero.

Diagonalizability of a Matrix:

A matrix A is diagonalizable if there exists a matrix P such that:

$$P^{-1}AP = D,$$

where D is a diagonal matrix containing the eigenvalues of A . A sufficient condition for diagonalizability is that the eigenvalues are **distinct**.

Step 2: Analyze the First Assumption

(a) The Matrix is Invertible:

- The eigenvalues of A are all distinct and real.
- For A to be invertible, $\det(A) \neq 0$. The determinant is the product of the eigenvalues:

$$\det(A) = \lambda_1 \cdot \lambda_2 \cdot \dots \cdot \lambda_n.$$

- Since the roots are distinct, none of the eigenvalues can be zero (if one eigenvalue were zero, the characteristic polynomial would not have distinct roots).

Conclusion: The matrix A is invertible because all eigenvalues are nonzero.

Step 3: Analyze the Second Assumption

(b) The Matrix is Diagonalizable:

- A sufficient condition for diagonalizability is that the matrix A has n linearly independent eigenvectors (for an $n \times n$ matrix).
- If all eigenvalues are distinct, it is guaranteed that there will be n linearly independent eigenvectors corresponding to these eigenvalues.
- Hence, A is diagonalizable because its eigenvalues are distinct.

Conclusion: The matrix A is diagonalizable because distinct eigenvalues guarantee n linearly independent eigenvectors.

Step 4: Final Conclusion

The data scientist is justified in making both assumptions:

1. The matrix is **invertible**, as all eigenvalues are distinct and nonzero.
2. The matrix is **diagonalizable**, as distinct eigenvalues guarantee the existence of n linearly independent eigenvectors.

